

MINTS: Nucleotide correspondence and control feasibility in genomic transformer interventions

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Abstract

Interventions in genomic transformers can compare different nucleotide positions even when their activation tensors have equal shapes. MINTS audits this failure and implements explicit nucleotide-span and vocabulary correspondence for variable-length tokenization, with matched sequence controls and separate targets for trained readouts and native predictions. Six of ten historical TATA patching pairs fail correspondence. Repaired CTCF patching of a discovery-selected head changes a fixed trained readout by 0.0157 decision-score units on 64 held-out sequences in 45 genomic clusters, with a 95% block interval of [0.0066, 0.0251], while its mean base density is near the background value of one. A separately frozen mapped-query natural-variant study retains 61 cases in 60 genomic clusters. Variant and sham Jensen–Shannon divergences average 0.013053 and 0.029018 nats; the paired contrast is -0.015966, with interval [-0.024454, -0.009146]. Larger sham shifts contradict the positive differential-sensitivity hypothesis on this small eligible population and stop native head search. We formalize two limited control-feasibility obstructions and report the eligibility gained by query correspondence without claiming that it isolates segmentation effects. The contribution is a reproducible computational validity audit, not evidence of biological binding causality or generally calibrated sensitivity to learned mechanisms.

Keywords: genomic transformers; nucleotide correspondence; activation patching; control feasibility.

1 Introduction

Genomic language models encode regulatory labels, but recovering a label from a representation does not identify the component that carries a motif-dependent computation. DNABERT attention visualizations can resemble known motifs [4]; activation patching can test component use, but its interpretation depends on corruption, controls and output metric [3, 10]. Variable-length DNA tokenization adds a concrete validity problem. Equal row indices can refer to different nucleotide intervals after an edit, so equal tensor dimensions alone do not establish a position-wise intervention.

MINTS addresses comparison of a specified prediction task across controlled edits when byte-pair encoding changes surrounding segmentation. It implements exact span-and-vocabulary query correspondence, validates channel-specific transfer and final-output restoration, and records control acceptance conditions. Compared with strict whole-input boundary matching, the adopted construction retains 61 matched scenarios where the original rules retain one of 12,288 screened scenarios. This gain establishes feasibility of matching prediction tasks, not unbiased isolation of motif effects. Limited proofs explain impossible controls under two constraint combinations. The contribution is a validity audit; algorithmic superiority or priority for alignment and component ablation are not established.

The inferential hierarchy separates fitted decoding, localization, intervention on a specified output and agreement with measured biology. A trained-readout effect and a failed native endpoint can coexist because they measure different computations. Distributed or query-specific motif information remains possible.

2 Related work and contribution scope

DART-Eval benchmarks regulatory feature discovery and counterfactual variant effects with predictive baselines [9]. MINTS adds component-level correspondence and control-feasibility checks rather than a broader benchmark. Global importance analysis measures population-level effects of embedded genomic patterns on predictions [5]; its input-level estimand differs from transferring internal head channels. Patching metrics and corruption choices already have a substantial methods literature [3, 10], so a patch effect is not itself a new method.

Ali’s July 2026 preprint already combines composition-matched binding comparisons, sparse dictionary ablation and native prediction-distribution interventions [2]. The inspected version removes dictionary directions and compares distribution shifts in bound and non-overlap windows. MINTS examines head contexts and verifies the nucleotide prediction task being compared. Native distribution changes establish effects within the model; biological binding interpretation requires independent measurement and provenance. MINTS does not claim the first composition-controlled native intervention.

LDARNet learns adaptive nucleotide boundaries and Motif-Vocab introduces statistically calibrated transcription-factor identity tokens [6, 7]. They change representation design rather than repair correspondence in a fixed BPE checkpoint. The present contribution is narrower: acceptance rules, falsifiable wiring checks, limited feasibility results and saved-evidence failures in one model. The audit evaluates validity and eligibility in a fixed checkpoint; it does not establish a general sensitivity guarantee or a new tokenizer design.

3 Methods

3.1 Model, data and targets

New computations use DNABERT-2 [11], pinned to revision `7bce263b15377fc15361f52cfab88f8b586abda0`. Native experiments load pretrained masked-language-model weights without missing or mismatched prediction parameters. Hook identity, repeat inference, attention reconstruction and final-query restoration check implementation, not biological labels or motif-specific sensitivity.

The accessible CTCF cohort uses GM12878 DNase ENCSR000EMT, GRCh38 file ENCF598KWZ. Cases overlap CTCF peaks from ENCSR000DKV, ENCF5827JRI; controls do not overlap those peaks. Matching uses chromosome, 204-base width, GC within 0.02 and log-DNase signal within 0.5. Non-overlap does not establish binding absence. JASPAR MA0139.1 is scanned in both orientations against a uniform log-odds background. Score-range fractions 0.7/0.8/0.9 are heuristic sensitivity choices, not calibrated local false-positive probabilities. Genome intervals are half-open GRCh38 coordinates; one-based archive positions are converted and reference alleles verified.

Promoter/splice contexts use the revised Nucleotide Transformer downstream release `851f9946252e90c665cdb3cc3eedb78f1f26197c`. Its promoter labels are attributed to EPDnew human release 006 and splice labels to GENCODE release 44, but not all sampling decisions are recoverable for independent reconstruction. Exact membership and label-source receipts accompany the supplement. Repaired head discovery uses chromosomes 18/19 and held-out evaluation 20/21; trained predictors use distinct training and validation populations.

3.2 Strict and mapped correspondence

For sequence x , tokenization returns rows i , spans $s_i(x) = [a_i, b_i)$ and vocabulary identities $v_i(x)$. Strict patching requires equal spans at every patched index; the controlled-edit construction additionally requires complete boundary agreement. Special rows are tracked separately. Six historical TATA pairs have equal tensor shapes but violate patched-position correspondence.

For reference, alternate and sham sequences $x^{(0)}, x^{(1)}, x^{(2)}$, mapped correspondence constructs

$$Q = \{(a, b, v) : \forall r \in \{0, 1, 2\}, \exists i_r \text{ with } s_{i_r}(x^{(r)}) = [a, b), v_{i_r}(x^{(r)}) = v\}.$$

Accepted queries have positive width and unchanged nucleotide content, lie outside motif/edit spans, and satisfy the frozen radius and same-side rules. The explicit map $\phi_r(a, b, v) = i_r$ allows indices to differ. Each query is masked separately, and the masked triplet must remain distinct so that masking cannot erase the

contrast. Checkpoint, vocabulary and unpadded hook interface must match. Twenty-nine of 61 scored cases shift at least one query index.

For head h , layer ℓ and head width d_h , transfer replaces only

$$C_\ell^{(r)}[\phi_r(q), hd_h : (h+1)d_h] \leftarrow C_\ell^{(0)}[\phi_0(q), hd_h : (h+1)d_h].$$

It does not replace an entire equal-index tensor. Final-query restoration replaces the full final row. The loaded pointwise prediction head must then reproduce the donor vocabulary logits within numerical tolerance. This is a wiring consequence, not proof of a head’s motif mechanism. Natural-variant head search never runs after the sensitivity gate fails, so its reported divergences are unpatched native predictions.

Executable correspondence certificate. The independent checker validates that real token spans form ordered complete partitions of each equal-length canonical input. It indexes span/identity tuples, merges motif intervals and sweeps reference spans. A candidate passes only if all three maps contain its interval and vocabulary ID, its nucleotide content is unchanged, geometry and exclusion rules hold, and masking leaves three distinct ID arrays. Unequal token counts and shifted indices are permitted. The checker certifies the declared maps and constraints; it cannot establish their tokenizer provenance or biological adequacy.

Soundness and completeness under declared maps. Assume a pinned vocabulary, complete ordered token partitions, an absent mask ID and fixed control inputs. The checker returns exactly the reference intervals satisfying the declared predicates. Partition uniqueness defines each ϕ_r ; the merged-interval sweep preserves the motif union. Unequal-length ID arrays remain distinct after masking. Equal-length arrays masked at different indices retain a mask/non-mask difference; masking at the same index removes only its ID difference. The pairwise difference test therefore equals literal masked-input comparison. Every satisfying interval passes the checks. This proves contract correctness, without identifying a biological mechanism.

For three inputs, the checker takes expected $O(T_0 + T_1 + T_2 + L + H \log H)$ time after tokenization, where L is nucleotide length and H is the number of motif intervals. Content comparisons total at most L per input because reference query spans are disjoint. Pairwise ID differences are counted once rather than copying complete inputs per query. Memory is $O(T_0 + T_1 + T_2 + H + Q)$ including the Q returned queries.

The post-study saved-input audit reproduces all 438 recorded queries in 61 cases and 60 clusters, including 29 cases with shifted query indices and 15 with complete boundary stability. It checks correspondence without scoring a model or changing eligibility, endpoints or stopped gates. Search checks capped sham candidates and tokenizes candidate triplets, with motif rescoring and forward passes adding separate costs. Query matching leaves segmentation and positional context elsewhere free to differ; it does not isolate a pure substitution effect inside the representation.

3.3 Controls and limited feasibility results

Trained-readout motif edits preserve exact A/C/G/T counts. Shams match directed base-transition multisets, edit count, window width, overlapping BPE count and covered width, introduce no new threshold-passing motif-hit locations and occupy the nearest eligible non-motif window. These constraints do not preserve all dinucleotides, other motifs or higher-order context.

Two-base donor obstruction. Assume GT changes to TG and the width-two sham must have the same directed substitution multiset $\{G \rightarrow T, T \rightarrow G\}$. Its starting sequence contains one G and one T. A non-GT sham therefore starts as TG and ends as GT, creating the prohibited hit. No control satisfies all these constraints. Wider-context controls are not ruled out.

Same-side distance obstruction. Assume disjoint width- w edit windows lie on one side of a fixed target and distance is the gap from the nearest window edge to the target. Their nearest edges differ by at least w , so target-distance differences are at least w . With $w = 19$ and tolerance two bases, the constraints are impossible. Overlapping windows, opposite sides or other distance definitions fall outside the proof. For same-side single-base edits, the distance difference equals edit separation, so a 16-base tolerance restricts the nominal 64-base sham search to 16 bases.

Table 1: Accessible CTCF prediction on fixed held-out membership. Marginal sequence-bootstrap intervals condition on fitted classifiers; the paired difference uses blocks.

Predictor	AUROC	95% sequence interval	<i>n</i>
Motif/position/composition	0.9199	[0.8844, 0.9539]	256
Residual only	0.6699	[0.5997, 0.7368]	256
Baseline + residual	0.8745	[0.8312, 0.9161]	256

Natural-variant shams use the same SNV substitution and local trinucleotide, with motif-location/score constraints and tight nucleotide geometry. Shuffles would change the perturbation being compared. Complete boundary and reference edit-token-width matching were relaxed during sequence-only development before the mapped protocol was frozen. Looser-geometry alternatives were recorded but not scored, and the preceding zero-control studies remain stopped.

3.4 Estimands and inference

Repaired patching measures patched minus corrupted fixed decision score. Motif-minus-sham effects average candidate edits within sequence and then sequences. Resampling groups transitively merge shared one-megabase intervals and all clean, edited and sham exact/RC identities. Blocks preserve the sequence estimand but are dependence proxies, not donor counts or certificates against distant homology.

Discovery selects one of 144 heads and freezes its identity and scorer before held-out outcomes. Primary edit-only denoising is separate from eleven exploratory secondary scheme/direction tests. Block claims use Holm correction of block p-values over that secondary family; sequence-Holm is separately named. Sensitivity uses 144-head maximum-statistic correction within configuration and the saved overall Holm family across metrics, fractions, geometries, heads and both resampling summaries. The family is not narrowed after inspection.

Mapped queries measure full-vocabulary Jensen–Shannon divergence in natural-log units. Queries weight nucleotide width, cases average within genomic clusters, and the primary estimator weights 60 cluster means equally. Its 95% interval resamples clusters with 10,000 draws. This differs from sequence-weighted readout effects. Intervals condition on fixed classifiers or a fixed checkpoint and omit fitting variability. Observational block sign-flip tests assume exchangeability, not randomized biological treatment.

4 Results

4.1 Prediction and repaired intervention

Held-out accessible CTCF membership has 128 overlap cases and 128 non-overlap controls; motifs occur in 113/128 cases and 33/128 controls. The direct baseline is informative, as Table 1 shows. Concatenation changes AUROC by -0.0454, with paired 95% block interval [-0.0756, -0.0166] over 81 clusters. Shared regularization across small baseline and high-dimensional residual features, selected from three strengths, can impair estimation. This establishes underperformance of this fitted concatenation, not absence of extra representational information.

CTCF search retains 12/22 scanned discovery positives and 64/107 held-out positives; TATA retains 12/122 and 16/106. Capped scans are not full-population retention estimates. CTCF’s 135 edits average within 64 sequences. Its selected layer 5/head 8 changes the trained readout positively under primary controls, while TATA’s interval crosses zero, as Table 2 shows. Sham distance has median/max 36/129 bases for CTCF and 116/266 for TATA, leaving contextual differences despite exact transition and token-width matching.

Selected CTCF clean mean base density is 1.0016, content-only density 0.8932 and position-only density 1.0069, with edits averaged within sequence. Base and position-only densities are near the background value of one, while content-only density is lower. These summaries and the positive patch effect do not complete a motif-detector claim. Query-specific and distributed computations remain unresolved. A 0.0157 decision-score effect does not imply large probability restoration or native binding prediction.

Table 2: Fixed-head trained-readout intervention on held-out chromosomes. Effects are sequence-weighted motif-minus-sham decision-score changes; 95% intervals and primary p-values use blocks. Indices are zero-based.

Motif	Head	n	Clusters	Effect	95% block interval	p
TATA	0/8	16	15	0.0243	[-0.0338, 0.1014]	0.5772
CTCF	5/8	64	45	0.0157	[0.0066, 0.0251]	0.0028

Table 3: Mapped-query native sensitivity on 61 cases in 60 genomic proxy clusters. Equal-cluster means and paired contrast are in nats; the interval is a 95% cluster bootstrap. Biological QC is incomplete.

Endpoint	Estimate
Variant JS	0.013053
Sham JS	0.029018
Variant minus sham	-0.015966
95% contrast interval	[-0.024454, -0.009146]

4.2 Sensitivity and fixture scope

Six fraction/geometry configurations have base-density within-configuration discovery counts 0, 1, 7, 44, 12 and 55. All six overall-corrected counts are zero under the saved full family. Both counts appear in the supplement; no-discovery is not proof of absent localization. Engineered calibration achieves perfect detection and zero false positives within implemented fixture families, which directly supply oracle motif features to QK/V and choose an effective head per observation. It verifies those computations and wiring, not natural-data selection or sensitivity to learned mechanisms.

4.3 Native endpoints fail advancement

The masked-flank pilot retains 12 sequences in 11 clusters. Motif-minus-sham target log-probability loss is -0.0568 nats, with 95% cluster interval $[-0.2513, 0.0869]$. It fails the frozen positive-sensitivity gate and stops before head selection. Its interval is too wide for a precise null. Recovery utility and successful restoration controls do not repair motif-specific endpoint feasibility.

The separately frozen mapped-query study retains 61 cases in 60 clusters. Table 3 and Figure 1 show larger sham shifts. Every leave-one-cluster-out primary mean remains negative. The positive differential-sensitivity gate fails, stopping causal head search, intervention-based power and independent confirmation.

The ADAstra archive has 512,556 records [1, 8]; 12,288 hash-selected scenarios were screened, 433 were motif-eligible and 61 matched/scored. These are separate populations. Original control rules retain 0/4,096 original and 1/8,192 expansion scenarios. Query matching alone retains 11 overall; removing edit-token-width matching as well yields the 61. Fifteen scored triplets have complete stable boundaries and all fifteen contrasts are negative, a descriptive post-inspection stratum rather than a new confirmation test.

These are reference-plus-one-SNV scenarios, not verified individual haplotypes. Symmetric JS cannot identify binding direction, and matchable cases may be unrepresentative. Segmentation, ALiBi position, context sensitivity and control differences remain alternatives. Live provenance exposes 238 ChIP observations for 26 cases but only five components after joining shared experiments. Genotype/phase, donor independence, replicate input, mapping correction and dosage remain unverified. Missing measurements are unknown, not zero. No biological correlation or biological confirmation is claimed.

5 Discussion

The practical outcome is a decision procedure for a specified component claim. First establish nucleotide and vocabulary correspondence. Then check control eligibility and wiring independently of biological outcomes. A held-out effect on a fixed readout supports contribution to that output, conditional on the retained controls.

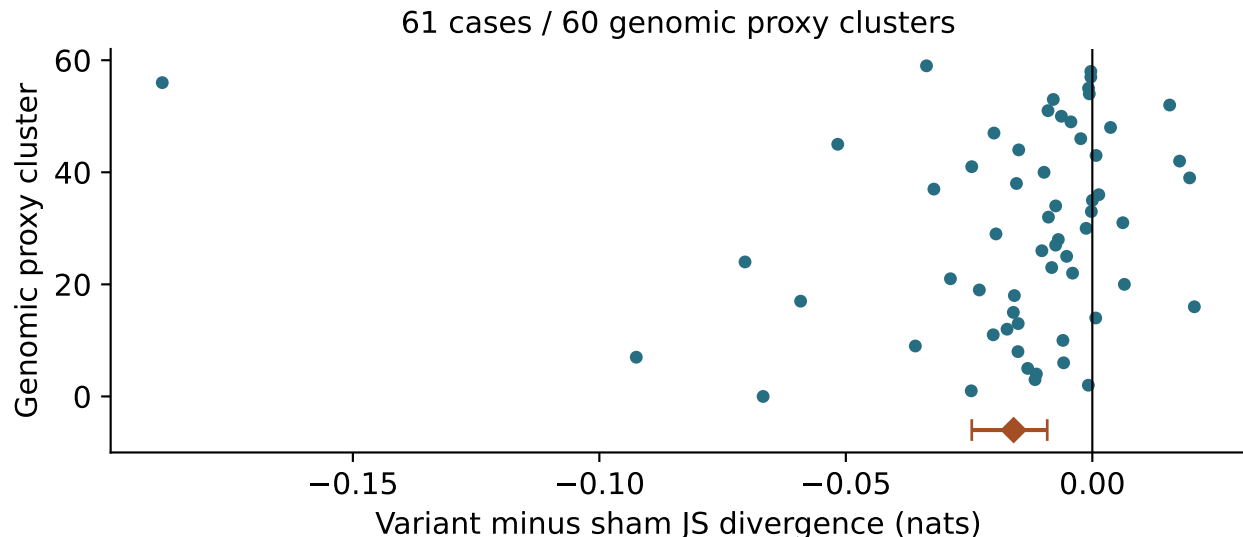


Figure 1: Cluster contrasts and the equal-cluster mean with 95% cluster-bootstrap interval in nats, conditional on fixed checkpoint and eligibility. Population: 61 cases/60 genomic dependence proxies.

Advancement to native head search requires its separately frozen endpoint to pass sensitivity. Agreement with biological binding requires independent biological measurements and provenance. These conditions are local acceptance rules for the implemented assays, not a validated universal evidence standard.

Table 4 separates conclusions that survive the audit from unresolved interpretations. A failed gate prevents the planned stronger claim; it does not establish that the model lacks motif information. The mapped-query result is useful because it exposes this distinction on saved natural-variant scenarios, even though it does not identify the source of the larger sham effects.

Table 4: Claim boundaries under the implemented evidence. Population sizes and uncertainty are reported with the corresponding result.

Evidence	Supported conclusion	Remaining limitation
Correspondence	Invalid equal-index transfers can be rejected; declared maps can be certified.	Tokenizer provenance and biological adequacy require separate checks.
CTCF readout	A fixed selected head contributes to the specified trained output under retained controls.	Native prediction and binding effects are not established.
TATA readout	The retained estimate is inconclusive.	The interval permits effects in either direction.
Native sensitivity	The pilot and mapped endpoint fail advancement.	Failed sensitivity does not imply absence of learned motif information.
Biological agreement	No confirmation is claimed.	Genotype, phase, independence and measurement QC remain incomplete.

Exact/RC and short-Hamming checks address limited leakage, but chromosome holdout does not exclude shifted homology, indels, distant repeats or pretraining exposure. BED scores are descriptive peak scores, not occupancy; entropy/run length are not repeat annotations. Full balance diagnostics, label specifications and failure histories accompany the supplement. The excluded Nucleotide Transformer comparison lacks established pinned loading and attention equivalence; no cross-architecture conclusion is drawn.

Original protocols, memberships, outcomes, timestamps and stopped studies are unchanged. Reporting v1 adds named Holm columns and overall sensitivity counts without model reruns. Identity-aware clustering preserves 45 CTCF and 15 TATA blocks. The source bundle links original receipts to revision `cc09923652f0b361f73a36f382c4358dcf8b8100` and corrected generator hashes. Compact predictions, cases, query maps,

memberships, plot data and 61 raw-logit files permit offline numerical reproduction without model/genome downloads.

Public code, the main paper, supplement and build inputs are at <https://github.com/ArjunCode/ss/MINTS>. Sequence-free reproduction data, including all 61 unchanged native logit files, are archived at <https://doi.org/10.5281/zenodo.23267982>. Run `python tools/verify_deposit.py` with `-data-root` pointing to the extracted dataset to verify file hashes, headline estimates, uncertainty and multiplicity corrections without model downloads. This uses saved cluster assignments; independent sequence reconstruction and fresh inference require upstream resources. Original code is MIT and original scientific outputs are CC-BY-4.0; third-party resources retain their upstream terms. No checkpoint weights, input DNA windows, full genomes or biological reads are included in the deposit.

6 Conclusion

MINTS establishes explicit nucleotide correspondence and diagnoses two limited feasibility failures. Equal shapes can conceal invalid interventions; repaired CTCF patching supports a specified trained-readout component effect, while TATA is inconclusive and the selected CTCF head’s mean base/position densities are near background under the measured statistic. Larger shams in the 61-case mapped-query study and all-negative leave-cluster-out means contradict positive differential sensitivity on its small eligible population. Failed gates remain failed, and incomplete biological QC prevents binding claims. The bounded validity audit is reproducible, while general sensitivity to learned mechanisms, cross-model transfer and biological binding interpretation remain unestablished.

Author declarations

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